

ORIGINAL ARTICLE

Platelet-mediated erythromelalgic, cerebral, ocular and coronary microvascular ischemic and thrombotic manifestations in patients with essential thrombocythemia and polycythemia vera: A distinct aspirin-responsive and coumadin-resistant arterial thrombophilia

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Abstract

Microvascular circulation disturbances including erythromelalgia, its microvascular ischemic complications, and migraine-like atypical or typical transient ischemic cerebral, ocular, and coronary ischemic attacks are specific clinical manifestations in patients with essential thrombocythemia (ET), and polycythemia vera (PV) associated with thrombocythemia. Thrombocythemia (ET and PV) patients with microvascular disturbances have shortened platelet survival, increased β -thromboglobulin (β -tg), platelet factor 4 (PF4), and thrombomodulin (TM) levels, and increased urinary thromboxane B2 (TxB2) excretion indicating platelet-mediated processes *in vivo*. Inhibition of platelet cyclooxygenase (COX 1) by aspirin is followed by relief of microvascular disturbances, correction of shortened platelet survival, and return of plasma levels of β -tg, PF4, TM levels and TxB2 excretion to normal. The transient ischemic attacks and thrombotic complications in thrombocythemia are very likely caused by hypersensitive platelets produced by spontaneously proliferating enlarged megakaryocytes in the bone marrow of ET and PV patients. In contrast to normal platelets in healthy individuals the circulating hypersensitive thrombocythemic platelets spontaneously activate and secrete their products, thus forming aggregates that transiently plug the microcirculation, or result in occlusive platelet thrombi in arterioles or small arteries. Clear evidence is presented that the microvascular transient ischemic and occlusive thrombotic complications in thrombocythemia patients are relieved by treatment with aspirin and by reduction of platelet counts to normal ($<400 \times 10^9/l$), but not by coumadin. In patients with thrombocythemia associated with PV, increased hematocrit and whole blood viscosity aggravate the platelet-mediated microvascular ischemic and thrombotic syndrome of thrombocythemia to major arterial and venous thrombotic complications. Correction of hematocrit and blood viscosity by phlebotomy significantly reduces the major arterial and venous thrombotic complications, but fails to prevent the platelet-mediated microvascular circulation disturbances in PV patients because thrombocythemia persists. Complete relief and prevention of microvascular and major thrombosis in PV patients are obtained by treatment with low-dose aspirin on top of phlebotomy or by treatment with the platelet lowering agents, anagrelide, interferon or hydroxyurea.

Keywords: Erythromelalgia, transient ischemic attacks, coronary artery disease, essential thrombocythemia, polycythemia vera, aspirin, coumadin

Introduction

Erythromelalgia (derived from the Greek words erythros = red, melos = extremity, and algos = pain) is a syndrome of redness and burning pain in the

extremities (Figure 1) [1]. Smith and Allen first noted that the relief of pain for several days after a single dose of aspirin (about 500 mg) is specific for erythromelalgia and can be used as a diagnostic criterion (Figure 1) [2]. In the 1970s evidence was



Figure 1. (A) Severe burning pain and red warm congestion of big and little toes, sole of the fore foot, and lateral edge of right foot, one week after discontinuation of aspirin (upper left and right). Burning pain and mottled red-blue ischemic discoloration of the sole of right foot, 2 weeks after discontinuation of aspirin (upper right). Painful swollen right fore foot sole (middle left) with acrocyanotic big toe (middle right). Progression of painful toes to black big toe and peeling of the skin of second and third toe in a 6-week period during treatment with coumarin (lower left) showed complete restoration of painful ischemic toes within a few weeks of treatment with aspirin (lower middle), but subsequently relapsed to painful red and blue toes within 2 weeks after discontinuation of aspirin (lower right). The skin surface temperature on thermography always exceeded 31°C (yellow, upper middle and central) at areas of red painful congestion (erythromelalgia) and was less than 25°C (light blue on thermogram, central and upper middle) in an acrocyanotic big toe (dark blue big toe, left and right middle). (B) Skin punch biopsies from skin areas of recently relapsed erythromelalgia in ET patients show arterioles with swollen endothelial cells and fibromuscular intimal proliferation (left). Skin punch biopsies from blue spots of skin areas with untreated erythromelalgia show arteriole with fibromuscular intimal proliferation with no thrombus, partial occlusion of the lumen and complete occlusion of arteriole by fresh or organized thrombus (middle). Biopsy taken from erythromelalgic skin areas showing an arteriole occluded by a thrombus, which showed a strong staining for von Willebrand factor (upper right) and a weak staining for fibrin (middle right) indicating a platelet-rich thrombus. Positive control fibrin staining in a patient with lupus anticoagulant and recurrent arterial thrombosis (lower right).

accumulating that aspirin sensitive erythromelalgia and its ischemic complications are related to the chronic myeloproliferative disorders essential thrombocythemia (ET) and polycythemia vera (PV) [3–9]. Erythromelalgia is initially felt as an itching and prickling sensation, often like pins and needles, which in time progresses to burning pain, red congestion, and acrocyanosis, provoked particularly by exercise or warmth (Figure 1) [3, 4]. In several cases the discomfort aggravates to such a degree that the patient has to sit with his feet elevated. Erythromelalgia may be located in the skin of the lower or upper leg or arm as red painful hot spots in the skin, frequently misdiagnosed as superficial

thrombophlebitis. Erythromelalgia may manifest in one or two or more extremities, but when bilateral it is always asymmetric (Figure 1). The sole of the fore foot, or one or more toes are more frequently involved than the palm of the hand, one or more fingers or fingertips. If left untreated acrocyanosis, necrosis of a nail fold or gangrene of a toe or finger tip or a whole distal toe phalanx may occur. Clinical studies showed that the broad spectrum of erythromelalgia, its microvascular complications, migraine-like, atypical or typical cerebral, and ocular ischemic attacks (TIA) and acute coronary syndromes in thrombocythemia vera are the consequence of one common pathway of platelet-mediated transient and

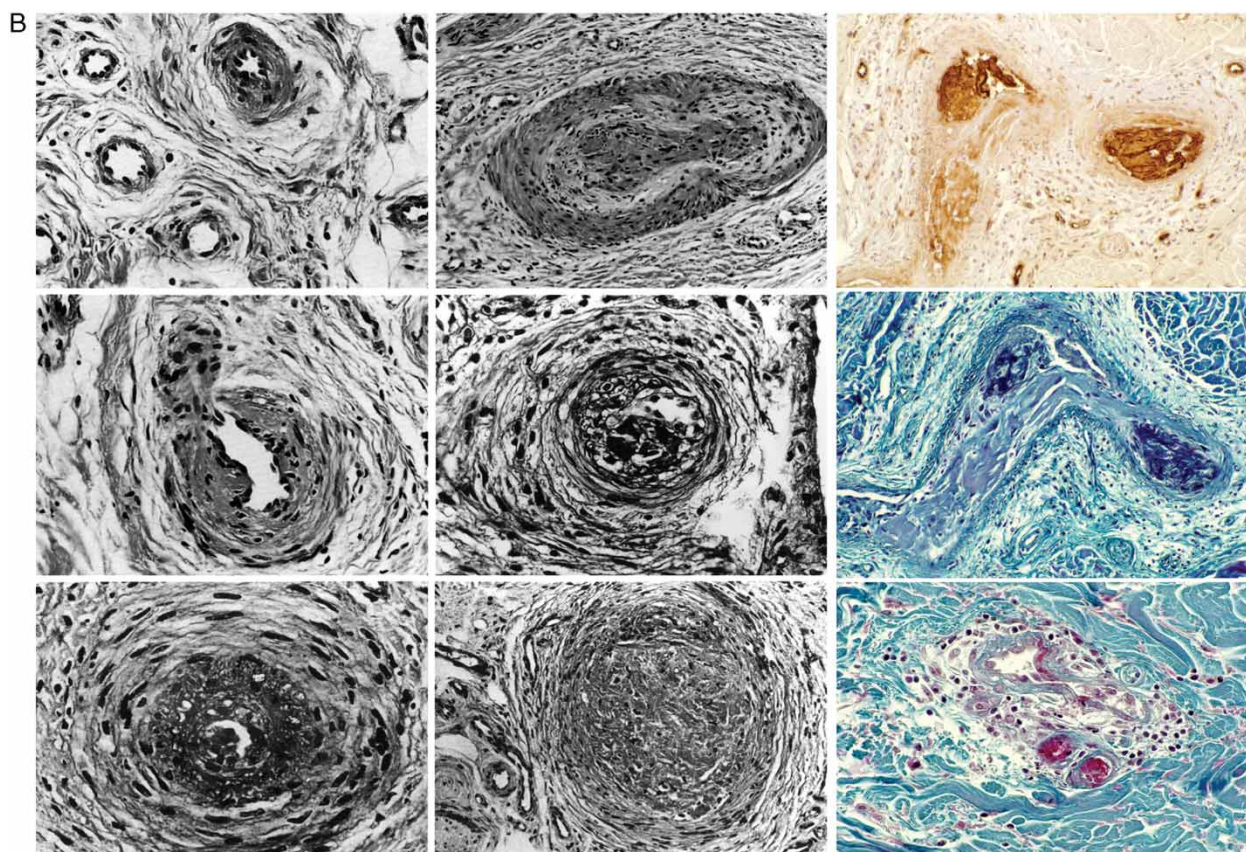


Figure 1. Continued.

ischemic events or thrombotic processes in the end-arterial microvasculature as the causative denominator, which can be prevented by aspirin through irreversible inhibition of platelet cyclooxygenase [10–18]. In this document we present the results of original histopathological, platelet kinetic, hemostatic and clinical intervention studies that elucidate the etiology and pathophysiology of aspirin sensitive platelet-mediated microvascular thrombosis in patients with thrombocythemia vera (ET and PV).

Methods

Essential thrombocythemia (ET) and polycythemia vera (PV) were diagnosed according to the European Clinical and Pathological (ECP) criteria (Table I, Figure 2) [19, 20]. Thermography of erythromelalgic skin areas was performed with a Bofors Mark II camera (Karskoga, Sweden) that indirectly registers the skin surface temperature. Multiple skin punch biopsies 0.3 cm were taken from areas of untreated erythromelalgia and from areas of recently relapsed erythromelalgia within 1–2 weeks after discontinuation of aspirin [12]. Skin punch biopsies were fixed in 40% formaldehyde, dehydrated in alcohol and embedded for light microscopic, and immunohistochemical studies [12]. For histopathological studies

sections of 2–4 µm thickness were stained with hematoxylin azophyloxine. For immunofluorescence, to characterize the arteriolar lesions, skin biopsies were snap-frozen in liquid nitrogen and cryostat section were tested using antiserum against factor VIII and against human smooth muscle cells. To detect the presence of von Willebrand factor (VWF) and fibrin in arteriolar thrombi derived from erythromelalgic skin areas, sections of 2–4 µm were stained for VWF with a streptavidin–biotin complex method using HRP-conjugated antibodies to human VWF (Dakopatts, Dako, Denmark) [15]. Staining of thrombi for fibrin was performed as described by Lendrum [21]. For platelet kinetic studies, autologous platelets were labelled with sodium ^{51}Cr chromate according to the modified method of Aster and Jandl [22, 23]. Platelet survival was determined by measuring the disappearance of platelet bound radioactivity from blood after injection of autologous ^{51}Cr -labelled platelets. Platelet survival was analysed by comparing computer fitting to linear, exponential and multiple hit models. The mean platelet survival was calculated according to the multiple hit model [22]. The half life time ($t_{1/2}$) and maximal life span (MaLS) were expressed as the time in days at which 50 and 95% of the injected ^{51}Cr platelets had disappeared from the circulation [24]. Platelets were counted in EDTA blood samples using

Table I. European clinical and pathological (ECP) criteria for the diagnosis of essential thrombocythemia (ET), thrombocythemia associated with prefibrotic idiopathic myelofibrosis (CIMF) mimicking ET, and polycythemia vera (PV) [19, 20].

Clinical criteria ET	Pathological criteria ET and CIMF
A1 Persistent increase of platelet count: in excess of $400 \times 10^9/l$, grade I 400–1000 and grade II more than $1000 \times 10^9/l$. A2 Normal spleen or only minor splenomegaly on echogram. A3 Normal or increased LAP-score, normal ESR and increased mean platelet volume (MPV, some large platelets). A4 Spontaneous megakaryocyte colony formation (CFU-Meg). A5 No signs or cause of reactive thrombocytosis (RT). A6 No preceding or allied other subtype of MPD, CML or MDS and absence of the Philadelphia chromosome.	ET. Predominant proliferation of enlarged megakaryocytes with hyperlobulated nuclei and mature cytoplasm, lacking conspicuous cytological abnormalities. No proliferation or immaturity of granulopoiesis or erythropoiesis. No or only borderline increase in reticulin (Figure 2). Prefibrotic CIMF mimicking ET Megakaryocytic and granulocytic myeloproliferation and relative reduction of erythroid precursors. Abnormal clustering and increase in atypical giant to medium sized megakaryocytes containing clumsy (cloud-like) lobulated nuclei and definitive maturation defects. No or slight myelofibrosis, leuko-erythroblastosis and/or anemia (Figure 2).
Clinical criteria PV	Pathological criteria PV
A1 Increase red cell mass (RCM) or hb (male > 18.5 g/dl, female > 16.5 g/dl) and ht (male > 51%, female > 48%). Or normal or slight increase of hb (male < 18.5 g/dl, female < 16.5 g/dl) and ht (male < 51%, female < 48%). A2 Persistent increase of platelet count: grade I: 400–1500, grade II: > 1500. A3 Splenomegaly on palpation or on ultrasound or CT (> 11 cm length in diameter). A4 Granulocytes > $10 \times 10^9/l$ and/or raised LAP-score in the absence of fever or infection.	PV Increased cellularity with trilineage myeloproliferation (i.e. panmyelosis). Proliferation and clustering of small to giant (pleiomorphic) megakaryocytes (Figure 2). Absence of stainable iron. No pronounced inflammatory reaction (plasmacytosis, cellular debris). Absence bone marrow features consistent with congenital polycythemia and secondary erythrocytosis. Spontaneous erythroid colony formation. Low serum Epo level.

ET bone marrow plus A1 is consistent with ET and exclude RT.

CIMF bone marrow plus A1 is consistent with prefibrotic AMM or IMF mimicking ET.

PV bone marrow plus low serum EPO or any of A is consistent with early PV at normal or slightly increased hb and ht, and with overt PV at increased hb, ht and RCM.

the Platelet Analyzer 810 (Baker Instruments). The inhibitory effect of antiplatelet agents on platelet cyclooxygenase (COX-1) activity was evaluated by the level of malondialdehyde (MDA) production in platelet-rich plasma. MDA production by arachidonic acid-stimulated platelets in platelet-rich plasma was measured according to the methods of Smith et al. [25]. Thromboxane B2 (TxB2) levels were determined in serum and urine samples by radio-immunoassay without further purification steps [17]. Plasma β -thromboglobulin (β TG) and platelet factor 4 (PF4) were measured using an enzyme-linked immunosorbent assay kit (Diagnostic Stago) [15]. To allow for between-groups comparisons the plasma β -TG and PF4 values were divided by the whole blood platelet count, and expressed as the corrected β -TG value and corrected PF-4 value, respectively. Plasma prothrombin fragment 1 + 2 (F1 + 2) was measured by an enzyme-immunoassay (Enzygnost F1 + 2 micro, Behring, Marburg, Germany). Total degradation products of fibrin and fibrinogen (TDP) were determined in plasma with an ELISA (Fibriostika TDP Microelisa). Thrombomodulin (TM) was measured with an ELISA kit (Asserachrom, Diagnostica Stago). Analysis of variance, followed by Student–Newman–Keuls (SNK

test) was used to identify pair-wise significant differences.

Gender, age, diagnosis and clinical symptoms at time of skin punch biopsies, thermography and platelet kinetic studies in 10 E+ patients (E+ = ET patients with symptoms) with thrombocythemia vera are the following.

- (1) Male, 45 years, ET. Disabling burning pain in the feet, red painful swelling of the ball of the right foot (Figure 1, upper right) and localized mottled painful patches in the skin of the upper leg.
- (2) Male, 38 years, ET. Pins and needles sensations and disabling burning distress of the lower extremities and red warm swelling of forefoot and toes.
- (3) Female, 55 years, ET. Burning and pain in red swollen fifth toe and along the lateral edge of the left foot.
- (4) Male, 74 years, classic PV in maintained remission by phlebotomy. Severe burning pain and red bluish swelling of the toes with a necrotic spot in the top of the big toe.
- (5) Male, 74 years, classic PV in maintained remission by phlebotomy. Progression of

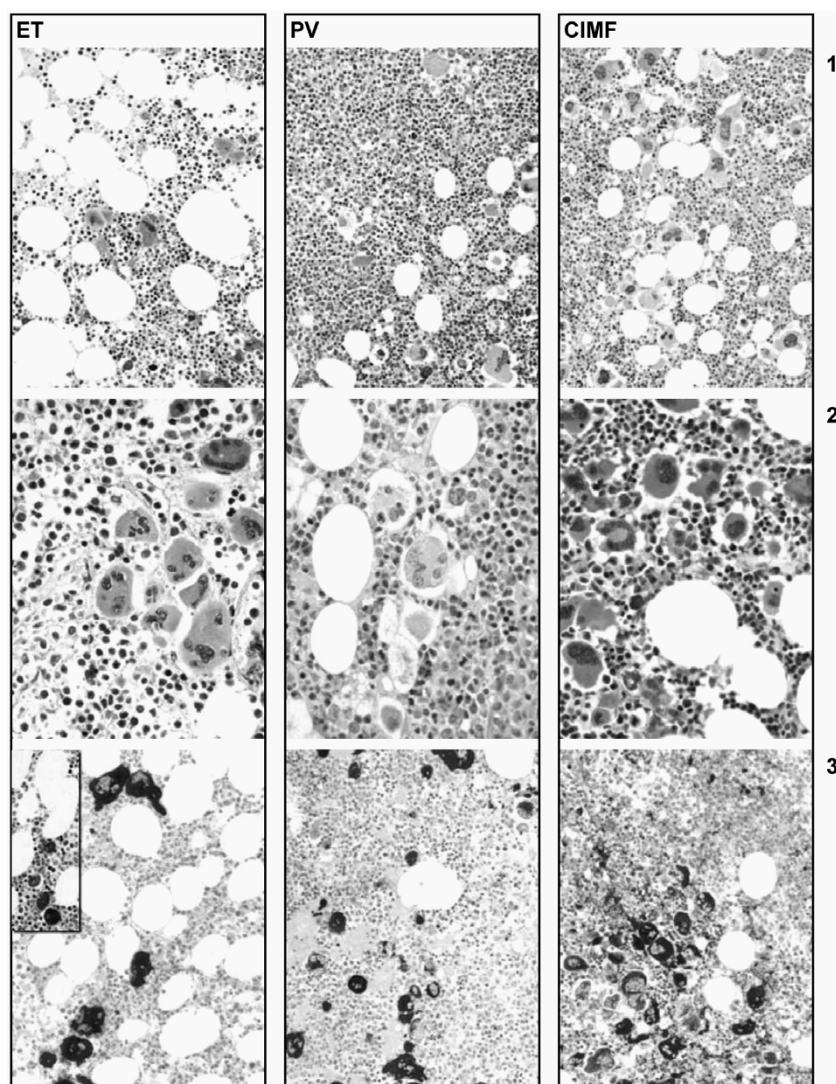


Figure 2. In contrast to normal sized megakaryocytes in reactive thrombocytosis and secondary erythrocytosis (inset), the bone marrow biopsy in thrombocythemia patients with various MPDs ET, PV and CIMF are featured by increase of clustered mature enlarged megakaryocytes with no increase of cellularity in ET (left), increase of clustered enlarged megakaryocytes with trilinear hypercellularity in PV (middle) and increase of dense clusters of enlarged immature megakaryocytes and hypercellularity due to dual megakaryopoiesis and granulopoiesis in CIMF (right). Megakaryocytes in ET are enlarged, with normal maturation and hyperploid nuclei in ET (left), megakaryocytes vary from small to giant (pleiomorphic) in PV (middle) and megakaryocytes show more or less dysmegakaryocytic features with atypical and immature so-called cloud-like nuclei in CIMF (right).

- burning, painful red fingers of the left hand to disabling painful swollen cyanosis. Multiple occlusions of small digital arteries on arteriogram.
- (6) Female 56 years, ET. Prickling pins and needles sensations, burning and red swollen fingers with episodic acrocyanosis.
- (7) Female, 61 years, classic PV in maintained remission by phlebotomy. Prickling sensations and painful red swelling of the big and small toe and the lateral edge of the right foot (Figure 1 upper left).
- (8) Male, 59 years, classic PV in maintained remission by phlebotomy. Burning distress and congestion of the left fore foot sole and painful swollen acrocyanotic first toe while on oral anticoagulation.
- (9) Male, 57 years, ET. Burning pain and warm dusky mottled redness of the right fore foot sole and cyanotic swollen and cyanotic painful cold big toe (Figure 1 middle left and right) while on oral anticoagulant treatment.
- (10) Female, 59 years, ET. Prickling sensations like pins and needles and burning pain in the right foot with slight acrocyanosis of the first, second and third toe.

Results and discussion

Erythromelalgia caused by platelet-mediated arteriolar thrombosis in thrombocythemia

The skin surface temperature of erythromelalgic areas in ET patients on thermography always

exceeded 31°C, and sometimes reached 36°C (yellow upper middle and central in Figure 1). Differences up to 9°C with the corresponding contralateral non-involved skin area were common. The skin surface of progressed erythromelgia to acrocyanosis of the big toe was much lower indicating end-arterial ischemic circulation disturbances (light blue coloured big toe central Figure 1).

The histopathological appearances from areas of recently relapsed erythromelgia 1–2 weeks after discontinuation of aspirin show arteriolar lesions in the reticular dermis without involvement of venules capillaries or nerves (Figure 1). The inner layer of the vessel wall was thickened by proliferation of cells, with narrowing of the lumen. Affected arterioles with proliferative thickening of its wall may be partially or completely occluded by organised thrombus (Figure 1). Immunofluorescence with factor VIII antisera on an affected arteriole shows an inner single layer of fluorescent endothelial cells, and with an antiserum against smooth muscle cells strong fluorescence was seen in the proliferated sub-endothelial cell zone and in the surrounding outer part of the media. The arteriolar thickening therefore is caused by proliferation of smooth muscle cells with increase of intercellular material: fibromuscular intimal proliferation. The secretion of platelet derived growth factor by *in vivo* activated platelet may be held responsible for the fibromuscular intimal proliferation of affected arterioles without or with occlusion of the lumen by platelet thrombi. Immunohistochemical studies of these biopsies revealed that thrombi in erythromelgic skin areas stained strongly with von Willebrand factor (Figure 1) opposed to only a weak staining for fibrin as compared to a positive staining for fibrin derived from a biopsy from a patient with a primary antiphospholipid syndrome and recurrent arterial thrombosis (Figure 1).

Aspirin is an irreversible inhibitor and indomethacin is a reversible inhibitor of platelet cyclooxygenase (COX1) activity [25–29]. Relief of erythromelgic pain from one oral aspirin (500 mg) occurred within 1 h and lasted 3–4 days, which is in accordance with the length of inhibition of platelet COX-1 activity as measured by malondialdehyde (MDA) production by arachidonic acid stimulated platelets in platelet-rich plasma (Figure 3). Relief of erythromelgic pain from one oral dose of indomethacin (75 mg) occurred within 1 h and lasted less than 1 day (24 h, Figure 3). Continued use of indomethacin 25 mg 3 times daily relieved erythromelgia and inhibited COX1 as measured by MDA in platelet-rich plasma. Erythromelgia appeared again within 1 day after discontinuation of indomethacin (Figure 3). Dipyridamole, sulfapyrazon, sodium salicylate and ticlopidine did not alleviate the erythromelgic pain and has no effect on platelet MDA production (Figure 3). Thus, active platelet COX-1 and

prostaglandin metabolism is necessary for erythromelgia to develop in patients with thrombocythemia vera.

Platelet kinetic studies were performed in 10 patients with thrombocytosis (group T, chronic myeloid leukemia in four, iron deficiency in two, Crohn's disease in one, malignancy in one and pulmonary abscesses in two), in 10 patients with thrombocythemia vera complicated by erythromelgia (group E+) and in 11 patients with asymptomatic thrombocythemia (group E–). Mean survival (MS), $t_{1/2}$, and maximal life span (MaLS) in the control group T were normal showing equal values for MS and MaLS with linear survival curves indicating a non-random, age-related disappearance of platelets from the circulation (Figure 4). The values for MS and $t_{1/2}$ in the E+ group were significantly shortened as compared to group T and group E– ($p < 0.01$). The MS was significantly shorter than the MaLS with pronounced curvilinear platelet survival curves indicating a consumptive disappearance of platelets from the circulation in group E+ (Figure 4). MS and $t_{1/2}$ were slightly but not significantly shortened as compared to group T (< 0.01). Platelet survival studies were repeated in seven E+ patients while on treatment with aspirin 500 mg/day which relieved the erythromelgic thrombotic symptoms completely and lastingly. The values of MS and $t_{1/2}$ before and after treatment were significantly different ($P < 0.05$) (Figure 5). The shortened values for MS and $t_{1/2}$ returned to normal, which was accompanied by a significant rise in platelet count (Figure 5). Two E+ patients while on adequate oral anticoagulant showed shortened MS and $t_{1/2}$ and a consumptive pattern of platelet disappearance, which reversed to normal values for MS and $t_{1/2}$ together with a change from a non-linear non-consumptive to a linear non-consumptive platelet disappearance curves by effective treatment of erythromelgia with aspirin. Dazoxiben, a selective inhibitor of platelet thromboxane synthetase activity, in two ET patients with erythromelgia (E+) did not alleviate the erythromelgic pain despite complete inhibition of platelet MDA and thromboxane B2 synthesis, and no inhibition of platelet prostaglandin endoperoxides [16].

Increased urinary TxB2 excretion has been described in ET and PV patients, probably reflecting spontaneous platelet activation in symptomatic ET (E+) patients [30, 31]. In three ET patients, who developed erythromelgia (E+) 4, 5 and 8 days after discontinuation of aspirin, we observed a progressive increase in urinary TxB2 excretion (Figure 6). At time of manifest erythromelgia a 3- to 30-fold (mean 7794 ± 3697 pg TxB2/mg creatinine) increase in urinary TxB2 levels was present (Figure 6). The urinary TxB2 excretion in three asymptomatic ET patients not using aspirin did exceed 900 pg TxB2 creatinine. Treatment of the three symptomatic ET

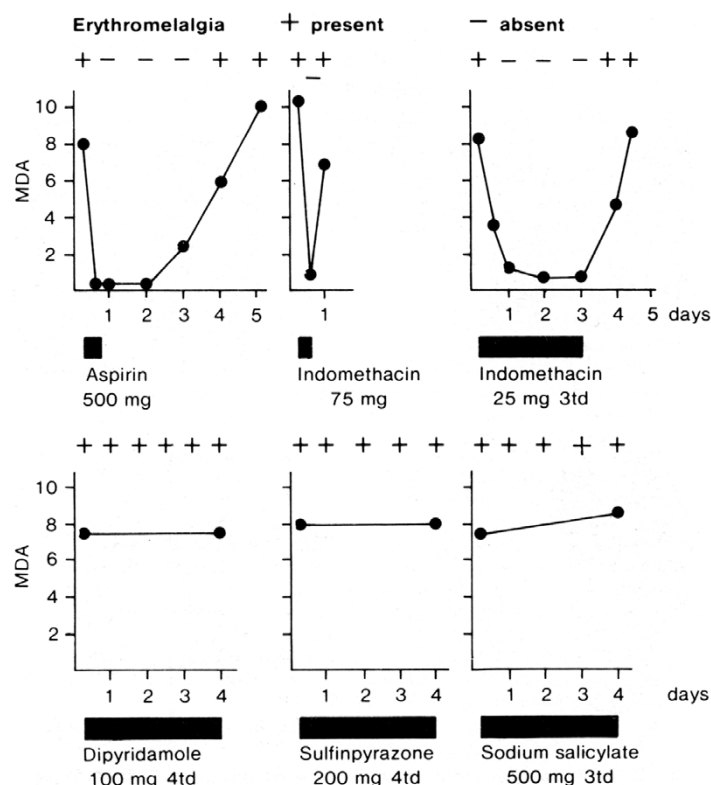


Figure 3. The effect of aspirin (acetylsalicylic acid), indomethacin, dipyridamole, sulfipyrazone and sodiumsalicylate on erythromelalgia and on platelet cyclooxygenase activity as measured by malondialdehyde (MDA) production by arachidonic acid-stimulated platelets in platelet-rich plasma of patients with essential thrombocythemia.

patients (E+) with low-dose aspirin 50 mg/day for 7 days resulted in a gradual decline in urinary TxB₂ excretion to near zero after 4–6 days in parallel with gradual relief of erythromelalgic distress and microvascular circulation disturbances (Figure 6). With a subsequent higher dose of aspirin (100 mg/day) at day 7, a further slight decrease of already suppressed urinary TxB₂ was observed (Figure 6).

The activation markers for endothelial cells (TM), platelets (PF₄ and β -TG) and coagulation (F1+2 and TDP) were measured in 20 healthy volunteers and in 21 ET patients, five with erythromelalgia (E+) and 16 without erythromelalgia (E-) (Figure 7) [15]. In comparison to control subjects and asymptomatic ET patients (E-), the presence of erythromelalgia in ET patients (E+) is associated with significantly higher values for TM (endothelial cell damage) and PF₄ and β -TG (activated platelets). A significant decrease of platelet activation markers (PF₄ and β -TG) and TM was obtained by curative treatment of erythromelalgia with aspirin ($P < 0.05$), which was accompanied by a significant rise of the platelet count (Figure 7). In contrast, there were no significant differences in the coagulation markers (F1+2 and TDP) in control E- and E+ patients (Figure 7).

The presented histopathological, platelet kinetic and intervention studies with aspirin show that erythromelalgia appears to be caused by intravascular

activation and aggregation of platelets with subsequent plugging or occlusion of the digital arterial microvasculature (Figures 1–7). These observations are in line with the concept that reversible platelet aggregates transiently plugging the microcirculation together with local release of their vasoactive and inflammatory secretion products may account for the vasomotor acroparesthesias and red painful congestion (erythromelalgia) as well as the migraine-like headaches frequently associated with atypical TIAs or followed by minor strokes in patients with thrombocythemia (Figure 8). The generation of thrombin appears not to be essential for the formation of microvascular platelet thrombi (Figure 7). These results give a plausible explanation for the inefficacy of coumadin and provide a rationale for using low-dose aspirin (40–100 mg/day, Figure 6) in the prevention and treatment of microvascular circulation disturbances in patients with thrombocythemia vera (Figure 8).

From 1974 to 1986 we performed a prospective study to evaluate the response to aspirin and busulfan in 20 symptomatic ET patients who presented with erythromelalgia, nine of them had a history of atypical or typical transient ischemic attacks (TIA, $n = 6$) or acute coronary syndrome ($n = 6$) (Figure 9) [18]. All ET patients responded to aspirin and became asymptomatic at platelet counts between 500 and $1500 \times 10^9/l$ (Figure 9). According to the

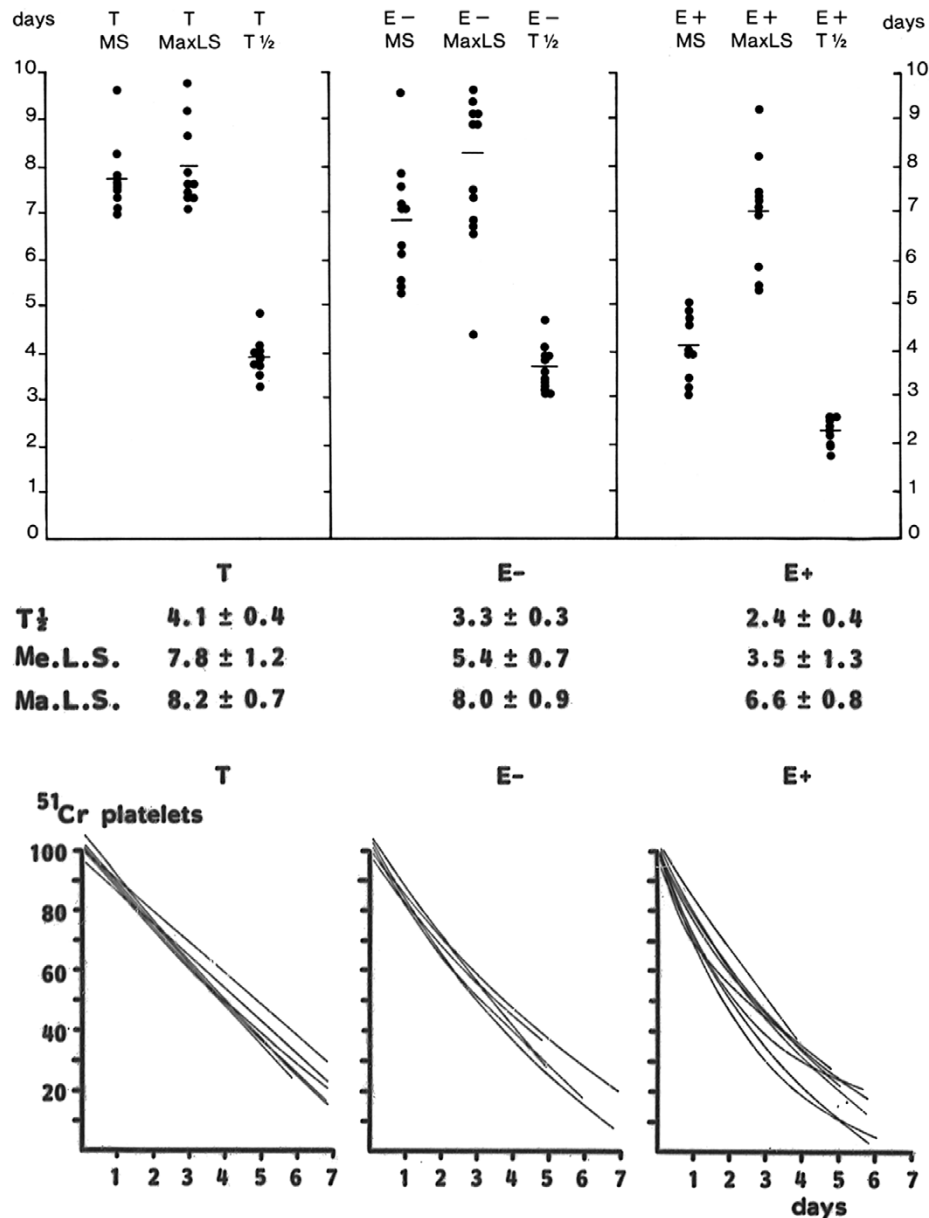


Figure 4. Platelet survival times and platelet disappearance curves in 10 ET patients complicated by erythromelalgia (E+), 10 asymptomatic ET patients (E-), and 11 asymptomatic patients with thrombocytosis (T). Curvilinear platelet survival curves in E+ indicates a consumptive disappearance of platelets from the circulation; a slight curvilinear platelet survival curves in E- suggests a slight, but insignificant random platelet consumption; and linear platelet survival curves in group T indicates a non-random, age-related disappearance of platelets from the circulation. MS=mean survival; $t_{1/2}$; MaLS=maximal life span.

protocol, ET patients were treated with one course of busulfan until a complete remission (platelet count $<350 \times 10^9/l$) was reached in all. At time of complete remission of ET, aspirin was discontinued and none of the ET patients became symptomatic as long as ET was in maintained remission for 4–61 months (mean 36 months, Figure 9). Eight ET patients became symptomatic (mainly erythromelalgia) at relapse of ET after 19–61 months at platelet counts between 410 and $635 \times 10^9/l$ (Figure 9). Four ET patients remained asymptomatic at relapse of ET after 31–46 months at platelet counts between 577 and $725 \times 10^9/l$.

The nature of neurological symptoms in 56 ET patients who presented with atypical transient ischemic attacks (TIA) and visual ischemic disturbances are described in detail by neurologists (Table II) [32]. The clinical manifestations and results of treatment in 30 ET patients with transient cerebral and visual symptoms or coronary artery disease are shown in Table III [33–35]. The transient focal and nonfocal neurological and visual symptoms all had a sudden onset, occurred with a march rather than all at one time, lasted for a few seconds to several minutes and were usually associated with or followed by a dull or pulsatile headache. This clinical

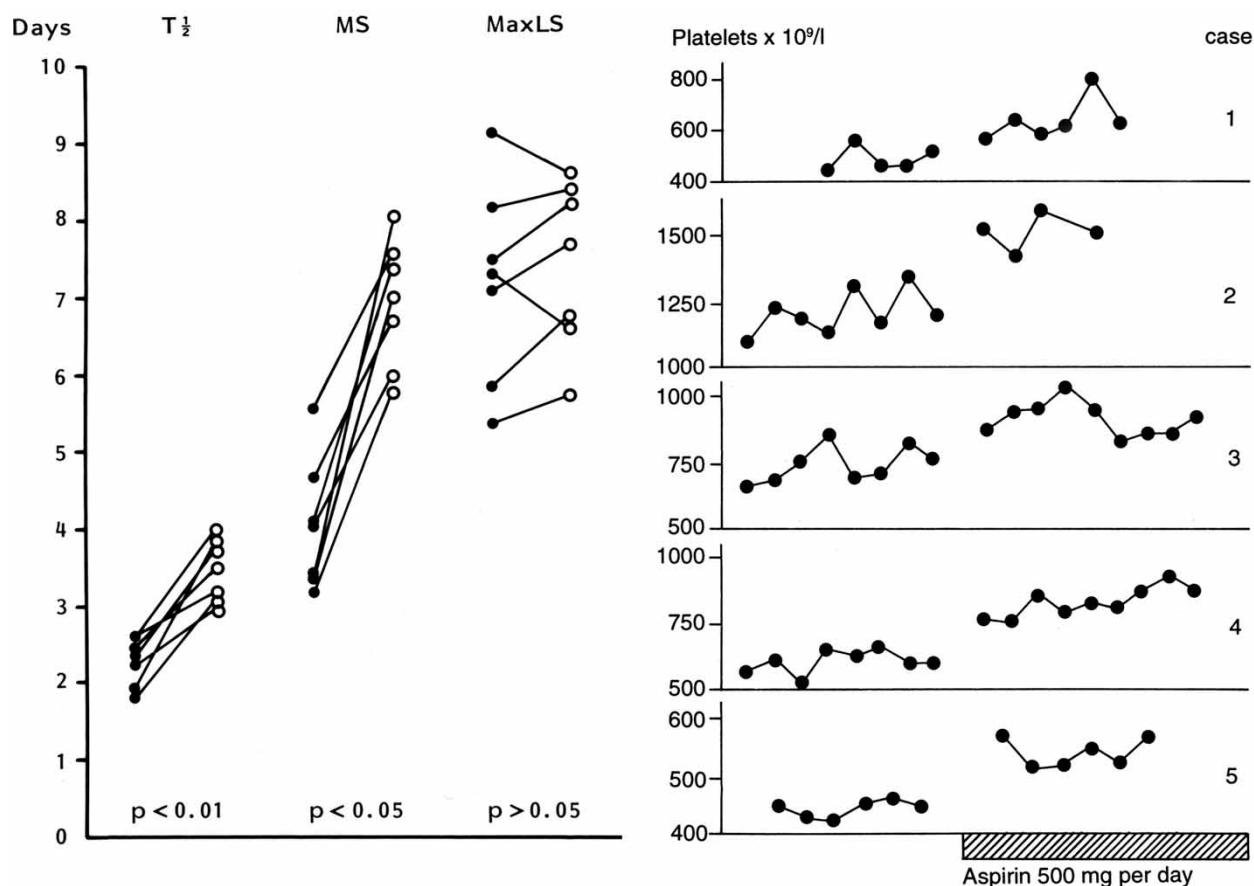


Figure 5. Platelet survival times in seven E+ patients before and after treatment with aspirin 500 mg/day, and platelet counts in four E+ patients before and after treatment with aspirin 500 mg/day.

presentation is very atypical for transient ischemic attacks caused by atherosclerosis, but the striking similarity with migraine accompaniments supports the crucial role of platelets in the pathogenesis of ischemic neurological disturbances in essential thrombocythemia (Tables II and III). This is in line with the concept that reversible platelet aggregates transiently plugging the microcirculation together with local release of their vasoactive and inflammatory secretion products may account for the aspirin-sensitive migraine-like headaches frequently associated with atypical TIAs or followed by minor strokes in patients with thrombocythemia (Figure 8). Microvascular circulation disturbances did occur in eight symptomatic ET patients while on adequate oral anticoagulation with coumadin (Table III). In this prospective observational study, ET patients who presented with cerebral ocular or coronary microvascular circulation disturbances were treated with aspirin followed by one course of busulfan. At time of complete remission aspirin was discontinued, but started again in as soon as platelet count rose to above $400 \times 10^9/l$ with the intention to prevent all types of microvascular disturbances. Thirteen ET patients in maintained complete remission of ET (platelet counts $< 400 \times 10^9/l$) for more than 1–6 years after treatment

with busulfan remained asymptomatic while not on aspirin at normal platelet counts (Figure 10). Twenty-three evaluable ET-patients with atypical TIAs remained asymptomatic while treated with aspirin (250–500 mg daily) for 1–9 years at increased platelet counts between 520 and $1380 \times 10^9/l$ (Figure 10). Fröhli documented the prophylactic efficacy of aspirin on vascular complications in 22 patients with ET ($n=9$) or thrombocythemia associated with PV in remission after bloodletting ($n=13$) at platelet counts between 400 and $1000 \times 10^9/l$ ³⁶. Aspirin (250 mg/day) in 22 symptomatic patients during a total period of 45 patient-years resulted in the disappearance of acral microvascular ischemia in 11, and of atypical TIAs including unstable gait, dizziness and blurred vision were completely abolished in 12 patients. Discontinuation of aspirin was followed by prompt recurrence of the microvascular circulation disturbances.

Efficacy and safety of aspirin in the prevention of thrombosis in ET and PV

The efficacy and safety of aspirin in the prevention of arterial thrombosis in ET was analyzed in our cohort of 68 ET patients diagnosed between January 1974

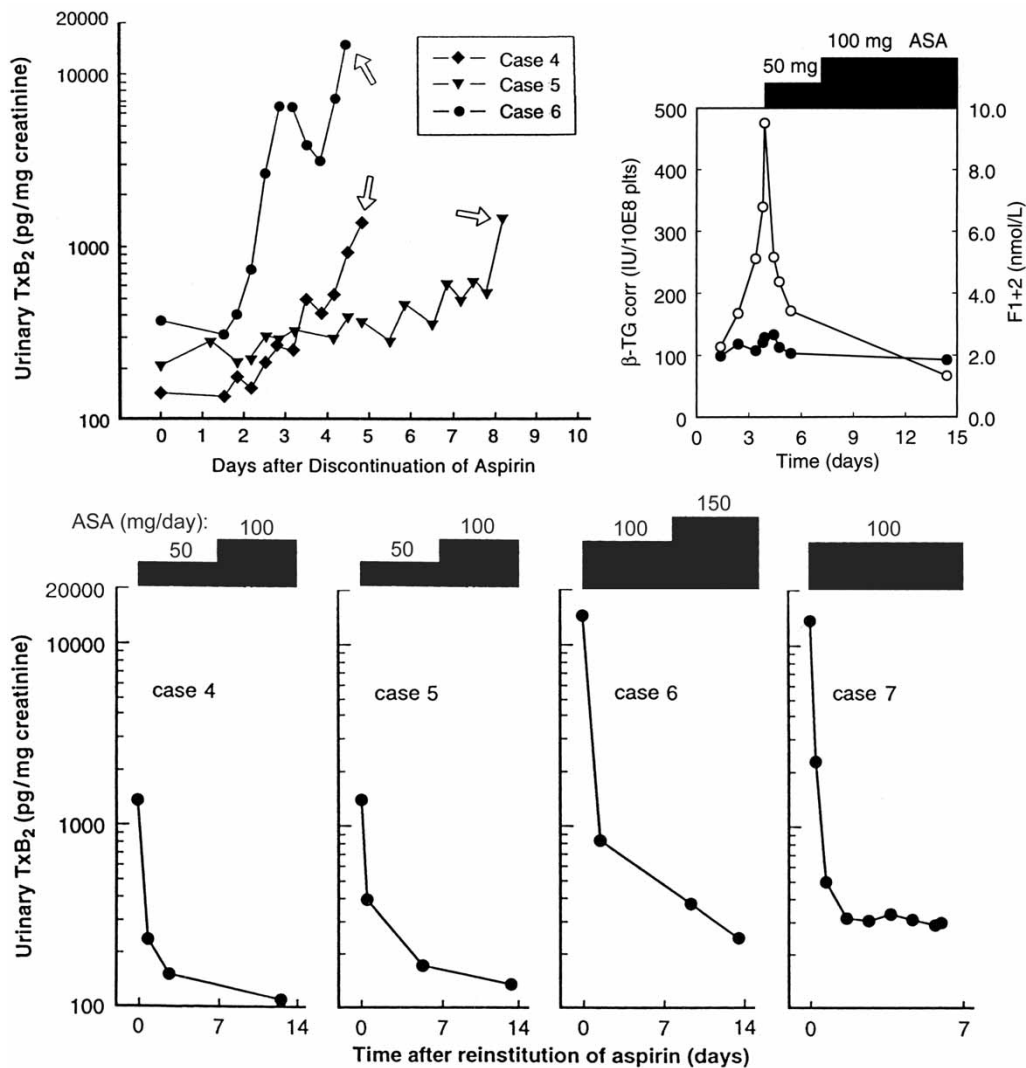


Figure 6. Temporal relationship between increasing urinary TxB₂ excretion and the development of erythromelalgia after discontinuation of aspirin (upper part). Temporal relationship of low-dose aspirin, 50 mg for 1 week, and inhibition of urinary TxB₂ excretion in four patients with essential thrombocythemia, who developed erythromelalgia within 4, 5 and 8 days after discontinuation and subsequent treatment with low-dose aspirin 50 mg/day gradually relieved erythromelalgia with a few days (lower part).

and December 1993 (mean follow-up 6.2 years (Table IV) [37, 38]. Sixty-eight consecutive ET patients were followed for a total of 419 patient years. At time of presentation 54 patients had ET-related microvascular thrombotic complications at platelet counts between 575 and 1031, mean $750 \times 10^9/l$, and treated with low-dose aspirin, a platelet lowering agent or both (Table IV, Figure 10). The 14 ET asymptomatic ET patients suffered from 27 thrombotic and two bleeding episodes occurred during a follow-up of 127 patient-years (=23% per 100 patient-years); platelet counts at time of thrombotic event were between 410 and 831, mean $610 \times 10^9/l$ (Table IV). Only four of the initially 14 asymptomatic ET patients remained asymptomatic after at least 30 person-years follow-up. In the treatment group of low-dose aspirin, five thrombotic events occurred during a follow-up of 179 patient years (2.8% per 100 pt/yrs). Ten bleeding episodes

occurred in ET patients receiving low-dose aspirin monotherapy during a follow-up of 139 patient-years (7.2% per 100 patient-years): platelet counts at time of bleeding were between 661 and $3460 \times 10^9/l$ (mean $1737 \times 10^9/l$) (Table IV, Figure 10). In patients receiving platelet lowering agents either busulfan or hydroxyurea and no aspirin, 10 thrombotic complications occurred during follow-up of 40 patient-years (25% per 100 patient-years): platelet count at time of thrombosis was $624 \pm 255 \times 10^9/l$, indicating inadequate control of thrombocytosis (Table IV, Figure 10). These data indicate that the incidence of thrombotic complications is high in ET patients who received no treatment, but was remarkably reduced in ET patients who received low-dose aspirin, or the combination of aspirin and cytoreductive treatment (Figure 10). ET patients receiving platelet lowering therapy and no aspirin the thrombotic risk persists when not reaching a

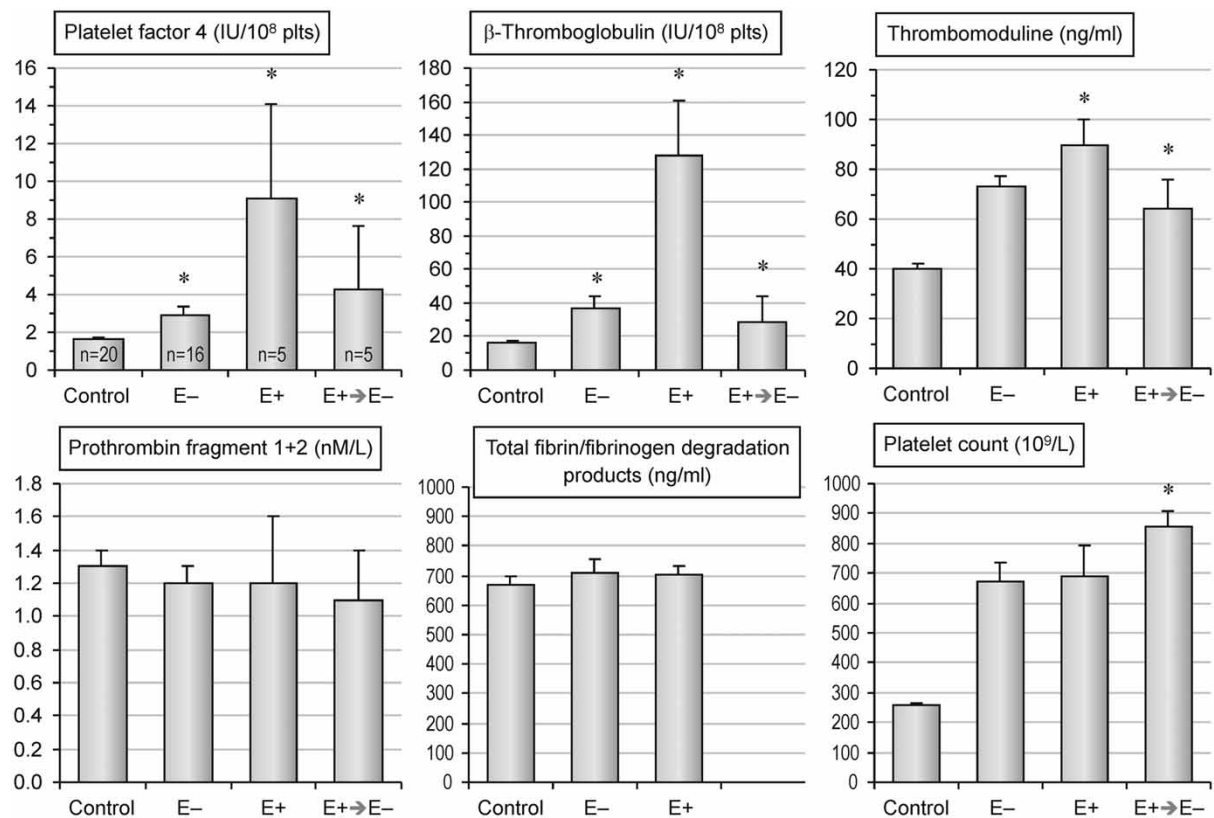


Figure 7. The effects of intervention with aspirin (500 mg/day) on hemostatic parameters in healthy controls, asymptomatic ET patients (E-), and ET patients suffering from erythromelalgia (E+). For explanation see text.

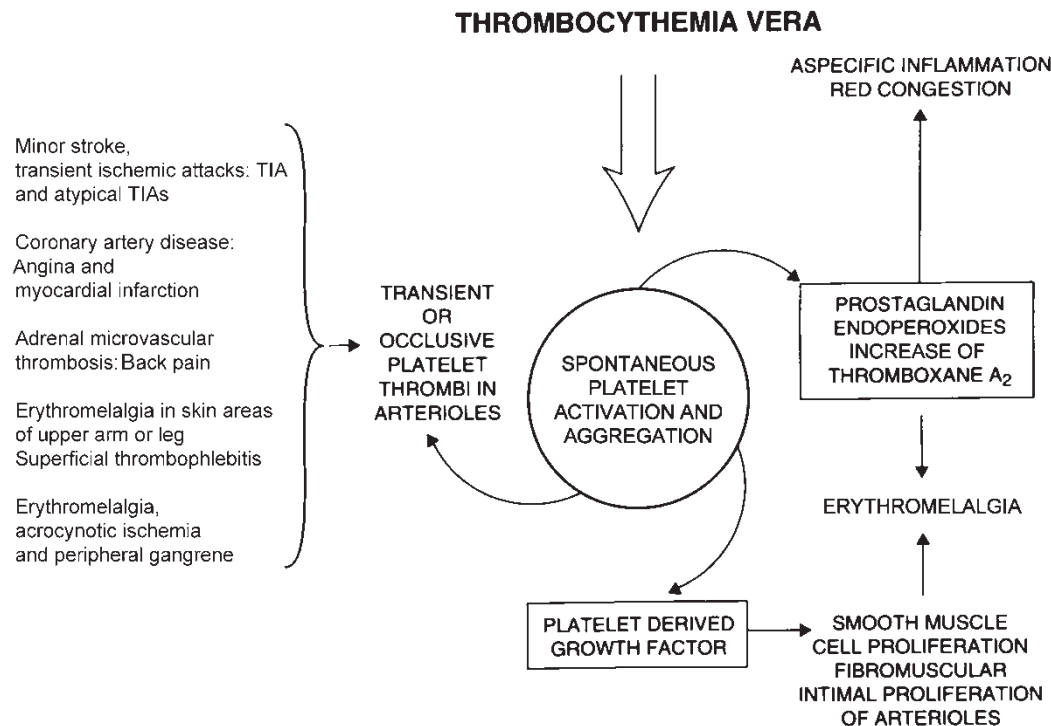


Figure 8. The broad clinical spectrum of microvascular circulation disturbances including erythromelalgia and its ischemic complications, atypical transient ischemic cerebral and ocular manifestations and/or coronary artery disease is caused by platelet cyclo-oxygenase-mediated ischemic and thrombotic processes in the end-arterial peripheral, cerebral and coronary circulation, which reflects a novel distinct disease entity of aspirin-responsive arterial thrombophilia in thrombocythemia vera.

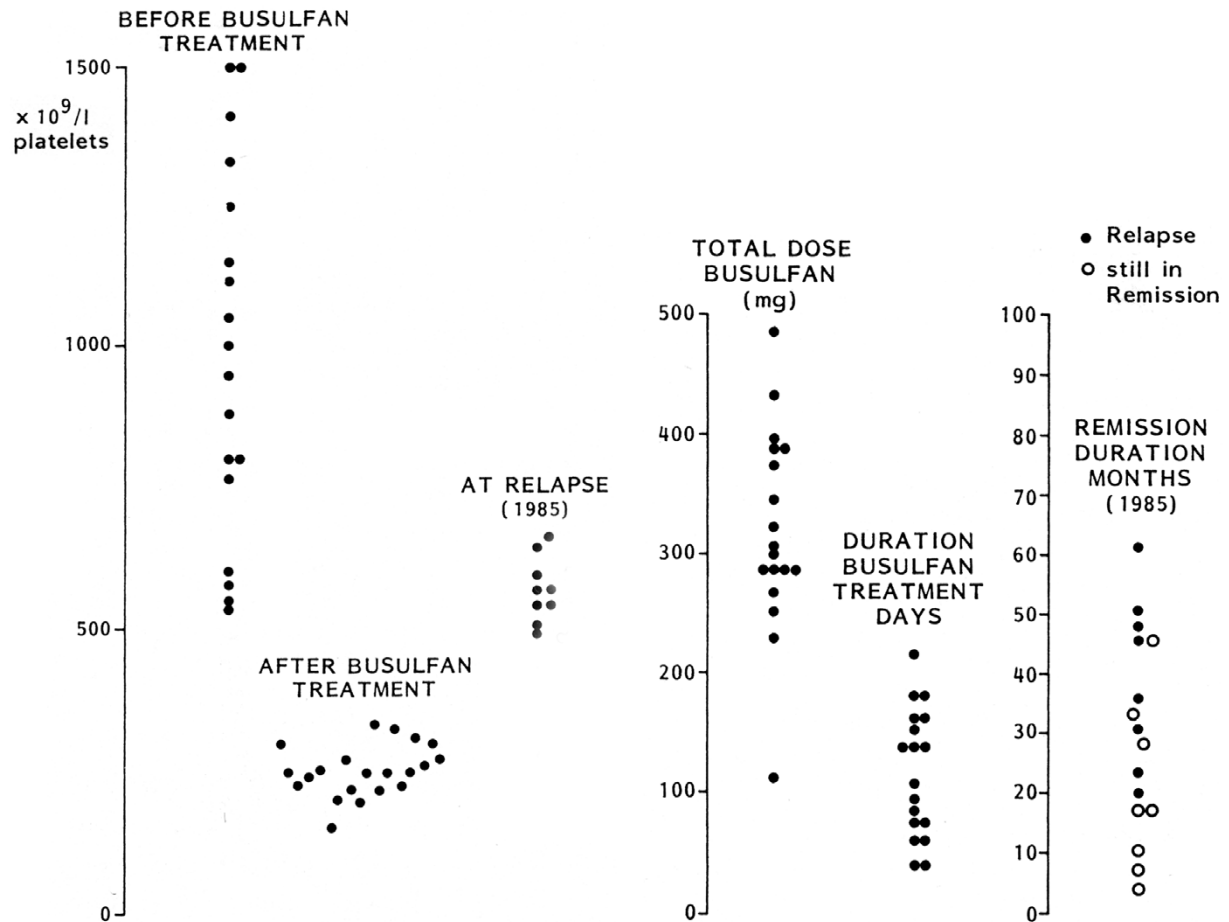


Figure 9. One course of busulfan in 20 symptomatic ET patients complicated by aspirin sensitive erythromelalgia uniformly induced complete remissions of thrombocythemia with no recurrence of erythromelalgia after discontinuation of aspirin as long as platelet counts remained normal. At time of relapsed thrombocythemia eight out of 12 ET patients again developed aspirin-sensitive erythromelalgia at platelet counts between 410 and 635 $\times 10^9/l$.

Table II. Nature of aspirin-responsive neurological and visual microvascular symptoms in 56 reported cases of essential thrombocythemia [32].

ET-related clinical manifestations	Number	Relative incidence
Acroparesthesias or numbness	13	24%
Painful toes and/or acrocyanosis (erythromelalgia)	12	21%
Transient ocular attacks:		
Visual scotomas	11	20%
Amaurosis fugax	5	9%
Diplopia	3	5%
Hemianopsia	2	4%
Blurred vision	14	25%
Transient ischemic attacks TIAs or hemiparesis arm and/or leg	17	30%
Atypical TIAs: Total	31	55%
Aphasia	3	5%
Dysarthria	12	21%
Unsteadiness or unstable gait	16	29%
Functional symptoms:		
Pulsatile headache	26	46%
Syncope	3	5%
Vertigo	2	4%
Dizziness	3	5%
Seizures	3	5%
Organic mental syndrome	1	2%

Table III. Focal and nonfocal transient ischemic attacks (TIAs) of cerebral and visual symptoms preceded or followed by erythromelalgia and/or coronary artery disease in 30 patients with essential thrombocythemia [33–35].

ET related clinical manifestations	Number of ET patients
Focal symptoms of TIAs	18
Transient monocular blindness	6
Transient mono- or hemiparesis	9
Both	4
Migraine accompaniments	3
Minor stroke	2
Nonfocal symptoms	19
Transient postural instability	14
Dysarthria	9
Scintillating scotomas	8
Erythromelalgia preceding or following TIAs	15
Coronary artery disease preceding or following TIAs	11
On Coumarin treatment at time of TIAs	8

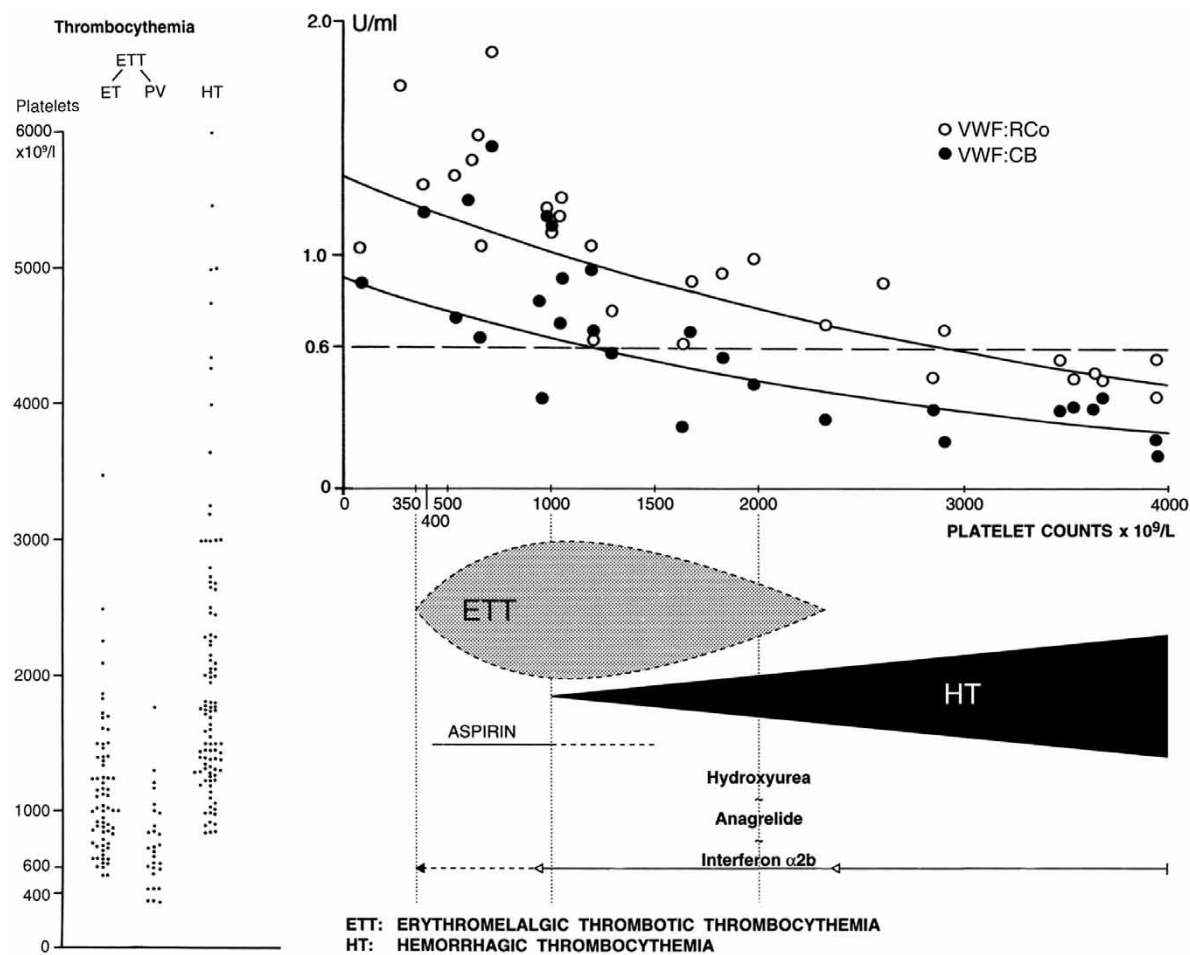


Figure 10. The Rotterdam concept on clinical and laboratory findings in erythromelagic, thrombotic thrombocythemia (ETT) and hemorrhagic thrombocythemia (HT) in patients with essential thrombocythemia or thrombocythemia associated with polycythemia vera in remission after bloodletting. Microvascular circulation ischemic disturbances events in thrombocythemia vera including erythromelalgia, atypical and typical cerebral and ocular transient ischemic attacks and acute coronary syndromes (ETT) already occur at platelet counts between 400 and 1000 $\times 10^9/l$. Low-dose aspirin is highly effective and safe in the cure and prevention of thrombotic and ischemic events and does not elicit bleedings at platelet counts below 1000 $\times 10^9/l$. At platelet counts between 1000 and 2000 $\times 10^9/l$, thrombosis and bleeding (ETT and HT) frequently occur in sequence or paradoxically and low-dose aspirin does prevent thrombotic complications but aggravates or may elicit bleeding symptoms. Reduction of the platelet count by anagrelide, interferon or hydroxyurea to below 1000 $\times 10^9/l$ by platelet lowering agents usually results in the disappearance of the bleeding tendency and improvement of the von Willebrand syndrome, but the thrombotic tendency persists as long as platelet counts are above the upper limit of normal (400 $\times 10^9/l$).

Table IV. Incidence of thrombotic and bleeding complications in 68 ET patients during long-term follow-up according to different treatment strategies [18, 37].

Treatment strategy	Duration of follow-up (person-years)	Thrombotic complications		Bleeding complications	
		Events (<i>n</i>)	Events/100 person-years	Events (<i>n</i>)	Events/100 person-years
<i>Asymptomatic 14 patients:</i>					
Watchful waiting	127	27*	33.3	2	1.6
<i>Symptomatic 54 patients:</i>					
Low-dose aspirin	139	5	3.6	10***	7.2
Platelet reduction	113	10**	8.9	2	1.8
Low-dose aspirin + platelet reduction	40	0	0.0	4	10.0
Total	419	42	10.0	18	4.3

Mean platelet count: *610, range $410\text{--}831 \times 10^9/\text{l}$ at time of thrombotic event (Figure 8); ** $624 \pm 255 \times 10^9/\text{l}$ at time of thrombotic event (Figure 8); ***1737, range $661\text{--}3460 \times 10^9/\text{l}$ at time of bleeding event (Figure 10).

Table V. Vascular complications, hematocrit and platelet counts in 65 patients with classical PV at time of presentation and after treatment with phlebotomy to correct hematocrit and busulfan for platelet reduction and not treated with aspirin [45].

	At presentation of PV	After treatment of PV
Number of PV patients	65	65
Follow-up	–	225 patient-years
Hematocrit	0.60 (0.49–0.74)	<0.48 before 1975 <0.45 since 1975
Platelet count $\times 10^9/\text{l}$ (mean)	512	390
Vascular occlusive events	32 patients = 49%	25 patients = 35%
<i>Microvascular/large vessels</i>	<i>13/19 (not specified)</i>	<i>Specified:</i>
Cerebrovascular events mainly TIA		15
Superficial thrombophlebitis or venous thrombosis		20
Microvascular disturbances		5
Myocardial infarction		1
Mesenteric vein thrombosis		1

Conclusion: Correction of hematocrit in PV to normal by bloodletting plus reduction of platelet count without the use of aspirin does not prevent vascular events (Figures 8 and 10).

compete remission (platelet count $<400 \times 10^9/\text{l}$) [37–39].

The large prospective randomized clinical trial of the polycythemia vera study group (PVSG-01) has overlooked the high frequency of microvascular circulation disturbances in patients with classical PV at time of diagnosis [40–44]. About two-thirds of PV patients in the PVSG 01 study had associated thrombocythemia [40]. At time of diagnosis in the PVSG 01 study, PV patients had increased red cell mass values between 36 and 66 ml/kg corresponding to high values for hematocrits between 0.50 and 0.75, resulting in hypervolumemia and increased whole blood hyperviscosity (Figure 8) [40], which was associated with a high frequency of major arterial and venous thrombosis, about 35% at time of PV diagnosis (Figure 12) [43, 44]. In the PVSG 01 study thrombosis free survival in treated PV patients aiming at a hematocrit below 0.50 was 50% for treatment with phlebotomy alone (no aspirin) and 25% for treatment with radioactive phosphor (^{32}P) during the first 5 year follow-up after initiation of

therapy [43, 44]. Despite the high thrombotic risk in the PVSG 01 study, the overall survival of phlebotomy alone arm was far superior to ^{32}P due to increased risk on leukemia and cancer [43, 44]. On top of the microvascular lesions typical of thrombocythemia, the incidence of major vascular episodes in PV correlated positively with increased hematocrit levels in a retrospective study of 65 PV patients (Table V, Figure 8) [45]. At time of diagnosis these 65 PV patients had a mean hematocrit of 0.60 and a mean platelet count of $512 \times 10^9/\text{l}$, and presented microvascular ischemic events and major thrombotic complication in 49% (Table V, Figure 8) [45]. This cohort of patients was treated with phlebotomy to correct hematocrit and busulfan to correct platelet count around $400 \times 10^9/\text{l}$, but no aspirin was given (Table V). The follow-up of these 65 treated PV patients for a total of 225 patient-years was associated with the occurrence of 42 vascular events in 23 patients (35%, Table V). The vascular episodes occurred at hematocrits below 0.44 in seven and at hematocrits above 0.45 in 18 treated PV patients.

The risk of major vascular episodes in PV is highest at hematocrits above 0.50, about 50%, moderately increased at hematocrits between 0.45 and 0.50, and rather low but still increased at hematocrits between 0.40 and 0.45 when not on low-dose aspirin (Table V, Figure 8) [45–48], indicating the need to add low-dose aspirin in treated PV patients [49]. The European Collaboration on Low-dose Aspirin in Polycythemia Vera (ECLAP) [50] performed a randomized clinical trial of low-dose aspirin vs. placebo to address the question whether low-dose aspirin indeed is safe and superior to placebo in preventing microvascular disturbances, major arterial thrombosis, and vascular death in treated PV patients [50]. Out of 1630 PV patients (median age 65 years) included in the ECLAP study, 742 (46%) had a clear indication for low-dose aspirin, 271 (16%) had a contraindication for aspirin and 518 (32%) PV patients, in whom the indication of aspirin was uncertain, were randomly allocated to aspirin 100 mg/day or placebo in a double-blind controlled trial [50]. Treatment of PV in these 518 patients allocated to the trial was according to the generally accepted recommendation: hydroxyurea in 44%, busulfan in 1%, pipobroman in 5.4%, interferon in 4.2% and phlebotomy alone 28%, or as adjuvant in 72%. The randomized PV patients showed no differences of vascular risk factors (like hypertension, diabetes, hyperlipidemia, previous thrombosis, etc.) in the aspirin and the placebo group [50]. Mean follow-up was 2.7 years. Mean values were 0.45 for hematocrit and $330 \times 10^9/l$ for platelet count. Treatment with low-dose aspirin significantly reduced the overall risk of a combined end-point of microvascular and major vascular complications including cardiac death, non-fatal myocardial infarction and stroke and major venous thrombosis from 15.1 to 6.7%: absolute risk reduction 8.4%, number needed to treat is 12 to prevent one thrombotic event [50]. This significant reduction was seen very soon after randomization. Major total and gastrointestinal bleeding were slightly increased without reaching statistical significance [50]. The ECLAP data confirm the Rotterdam concept on platelet-mediated thrombosis in ET (Figure 8) and PV (Figure 8) that low-dose aspirin is safe and effective in the prevention of minor and major thrombosis in essential thrombocythemia and should be included on top of phlebotomy and/or treatment with hydroxyurea in PV patients, even when platelet count and hematocrit are corrected to normal.

The paradox of platelet-mediated thrombosis and bleeding in thrombocythemia vera

Clinicians should be aware that a starting low dose of aspirin, 50 mg daily, in symptomatic ET and PV patients complicated by erythromelalgia induces a slow relief of pain and gradual inhibition of platelet

cyclooxygenase (COX-1), as it takes 4–6 days to completely inhibit platelet COX-1 and to relief of erythromelalgia by such a low dose (Figure 6) [17, 27, 39]. Consequently, symptomatic thrombocythemia vera patients presenting with platelet-mediated microvascular circulation disturbances should be properly diagnosed and immediately treated with a loading dose of aspirin 300–500 mg followed by a low dose 50–80 mg daily. Thrombocythemia with platelet counts between 1000 and $2000 \times 10^9/l$ is usually complicated by minor and/or major thrombotic complications and frequently associated with the paradoxical occurrence of microvascular thrombosis complications and bleeding manifestations (Figure 10) [51–60]. In this situation aspirin will prevent the thrombotic complications, but will elicit or aggravate the bleeding tendency indicating the urgent need to lower platelet counts (Figure 10). Mucocutaneous bleedings spontaneously occur at increasing platelet count in excess of 1000 – $1500 \times 10^9/l$, which is associated with an acquired type II-like von Willebrand syndrome featured by low levels of functional von Willebrand factor (VWF) parameters due to the absence of high and intermediate VWF multimers (Figure 10) [51–58]. Reduction of platelet count by anagrelide, interferon or hydroxyurea to less than $1000 \times 10^9/l$ will result in the disappearance of the bleeding symptoms, improvement of VWF-ristocetine cofactor activity to low normal levels and significant improvement of the VWF multimeric pattern with reappearance of the intermediate and some of the large multimers, but with the absence of the largest vWF-multimers (Figure 10) [51–60]. Correction of the platelet counts to normal is associated with complete correction of the VWF-multimeric pattern and correction of all VWF-parameters to complete normal values.

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